Theory for the LASSO

I) Results assuming a linear model

- Consider the linear model $\underline{y} = X\beta + \varepsilon$ where $\varepsilon \sim \mathcal{N}(0, \sigma^2 I_m)$ and denote by β^0 the true value of the parameter. ($X \in \mathbb{R}^{m \times p}$)
- Then: $\hat{\boldsymbol{\beta}}^{\text{ls}} \sim \mathcal{N}\big(\beta^0, \sigma^2 (X'X)^{-1}\big)$

$$
\begin{aligned}
&\Rightarrow \sigma^{-1}(X'X)^{1/2}\left(\hat{\boldsymbol{\beta}}^{\text{ls}} - \beta^0\right) \sim \mathcal{N}(0, I_p) && (1)\\
&\Rightarrow \sigma^{-2}\left(\hat{\boldsymbol{\beta}}^{\text{ls}} - \beta^0\right)'(X'X)^{1/2}(X'X)^{1/2}\left(\hat{\boldsymbol{\beta}}^{\text{ls}} - \beta^0\right) \sim \chi^2(p) && (2)\\
&\iff \left(X\left(\hat{\boldsymbol{\beta}}^{\text{ls}} - \beta^0\right)\right)'\left(X\left(\hat{\boldsymbol{\beta}}^{\text{ls}} - \beta^0\right)\right)/\sigma^2 \sim \chi^2(p) && (3)\\
&\iff \frac{\left\|X\left(\hat{\boldsymbol{\beta}}^{\text{ls}} - \beta^0\right)\right\|_2^2}{\sigma^2} \sim \chi^2(p), \quad \text{hence:} && (4)
\end{aligned}
$$

$$
\mathbb{E}\left[\frac{\left\|X\left(\hat{\boldsymbol{\beta}}^{\text{ls}} - \beta^0\right)\right\|_2^2}{\sigma^2}\right] = p \qquad \big(\text{if } \sigma \sim \chi^2(p),\ \text{then } \mathbb{E}\sigma = p\big) \tag{5}
$$

$$
(*) \Rightarrow \frac{\mathbb{E}\left\|X\left(\hat{\boldsymbol{\beta}}^{\text{ls}} - \beta^0\right)\right\|_2^2}{m} = \frac{\sigma^2}{m}p \tag{6}
$$

$\Rightarrow$ After reparameterizing to an orthonormal design ($X'X = I_p$), each $\hat{\beta}_h^0$ is estimated with squared accuracy σ^2/m.

- Consider $p > m$ and assume the true model is sparse. Consider

$$
S_0 = \left\{j = 1, \ldots, p : \beta_j^0 \neq 0\right\} \quad \text{s.t.} \quad |S_0| = s_0 \tag{7}
$$

the active set of β^0 and the sparsity index, respectively.

- If we knew S_0, we would simply neglect variables $\{x_j\}_{j \notin S_0}$ and obtain an overall squared accuracy of $\sigma^2 s_0/m$.
- As S_0 is unknown, we use the ℓ_1 penalty and estimate:

$$
\hat{\boldsymbol{\beta}}^{\text{LASSO}} = \arg\min_{\beta} \frac{1}{m}\|\underline{y} - X\beta\|_2^2 + \lambda\|\beta\|_1 \tag{8}
$$

- We wish to investigate the properties of the LASSO, including accuracy.
- By definition of $\hat{\boldsymbol{\beta}}^{\text{LASSO}}$, we get:

$$\frac{\left\|\underline{y} - X\hat{\boldsymbol{\beta}}^{\text{LASSO}}\right\|_2^2}{m} + \lambda \left\|\hat{\boldsymbol{\beta}}^{\text{LASSO}}\right\|_1 \leq \frac{\left\|\underline{y} - X\beta^0\right\|_2^2}{m} + \lambda \left\|\beta^0\right\|_1 \tag{9}$$

$$\iff \frac{\left\|X\beta^0 + \varepsilon - X\hat{\boldsymbol{\beta}}^{\text{LASSO}}\right\|_2^2}{m} + \lambda \left\|\hat{\boldsymbol{\beta}}^{\text{LASSO}}\right\|_1 \leq \frac{\|\varepsilon\|_2^2}{m} + \lambda \left\|\beta^0\right\|_1 \tag{10}$$

$$\iff \frac{\|\varepsilon\|_2^2 + \left\|X\left(\hat{\boldsymbol{\beta}}^{\text{LASSO}} - \beta^0\right)\right\|_2^2 - 2\varepsilon' X\left(\hat{\boldsymbol{\beta}}^{\text{LASSO}} - \beta^0\right)}{m} + \lambda \left\|\hat{\boldsymbol{\beta}}^{\text{LASSO}}\right\|_1 \leq \frac{\|\varepsilon\|_2^2}{m} + \lambda \left\|\beta^0\right\|_1 \tag{11}$$

$$\iff \frac{\left\|X\left(\hat{\boldsymbol{\beta}}^{\text{LASSO}} - \beta^0\right)\right\|_2^2}{m} + \lambda \left\|\hat{\boldsymbol{\beta}}^{\text{LASSO}}\right\|_1 \leq \frac{2\varepsilon' X\left(\hat{\boldsymbol{\beta}}^{\text{LASSO}} - \beta^0\right)}{m} + \lambda \left\|\beta^0\right\|_1 \tag{$**$}$$

- Note that $X\left(\hat{\boldsymbol{\beta}}^{\text{LASSO}} - \beta^0\right) = \sum_{j=1}^{p} x_j \left(\hat{\beta}_j^{\text{LASSO}} - \beta_j^0\right)$

$$\Rightarrow \varepsilon' X\left(\hat{\boldsymbol{\beta}}^{\text{LASSO}} - \beta^0\right) = \sum_{j=1}^{p} \left(\hat{\beta}_j^{\text{LASSO}} - \beta_j^0\right) \varepsilon' x_j \tag{12}$$

$$\Rightarrow \left|\varepsilon' X\left(\hat{\boldsymbol{\beta}}^{\text{LASSO}} - \beta^0\right)\right| = \left|\sum_{j=1}^{p} \left(\hat{\beta}_j^{\text{LASSO}} - \beta_j^0\right) \varepsilon' x_j\right| \tag{13}$$

$$\leq \sum_{j=1}^{p} \left|\hat{\beta}_j^{\text{LASSO}} - \beta_j^0\right| \left|\varepsilon' x_j\right| \tag{14}$$

$$\leq \max_{j=1,\ldots,p} \left|\varepsilon' x_j\right| \sum_{j=1}^{p} \left|\hat{\beta}_j^{\text{LASSO}} - \beta_j^0\right| \tag{15}$$

$$\Rightarrow 2\left|\varepsilon' X\left(\hat{\boldsymbol{\beta}}^{\text{LASSO}} - \beta^0\right)\right| \leq \max_{1 \leq j \leq p} 2\left|\varepsilon' x_j\right| \; \left\|\hat{\boldsymbol{\beta}}^{\text{LASSO}} - \beta^0\right\|_1 \tag{$*$}$$

The point is that the penalty should overrule the empirical process $\dfrac{2\varepsilon' X\left(\hat{\boldsymbol{\beta}}^{\text{LASSO}} - \beta^0\right)}{m}$.

- Consider the set $T = \left\{ \max_{1 \le j \le p} 2|\varepsilon' x_j|/m \le \lambda_0 \right\}$
- For a suitable value of λ_0, T has "large" probability, so that the F.O.C.s of the LASSO turn into a purely deterministic problem.
- Denote by $\hat{\sigma}_j^2$ the diagonal elements of $X'X/m$, $\forall j = 1, \dots, p$.
- Suppose $\hat{\sigma}_j^2 = 1\ \forall j$, then, as (meaning that the data is standardized):

$$\varepsilon \sim \mathcal{N}(0, \sigma^2 I_m) \Rightarrow \varepsilon' x_j \sim \mathcal{N}\big(0, \sigma^2 x_j' x_j\big) \iff \varepsilon' x_j \sim \mathcal{N}\big(0, \sigma^2 \hat{\sigma}_j^2 m\big) \tag{16}$$

$$\iff \varepsilon' x_j \sim \mathcal{N}(0, \sigma^2 m) \iff \frac{\varepsilon' x_j}{\sqrt{m\sigma^2}} \sim \mathcal{N}(0,1), \text{ s.t.:} \tag{17}$$

$$\mathbb{P}\left(\max_{1 \le j \le p} \left| \varepsilon' x_j / \sqrt{m\sigma^2} \right| > \sqrt{t^2 + 2\log p} \right) \le \sum_{j=1}^{p} \mathbb{P}\left(\left| \varepsilon' x_j / \sqrt{m\sigma^2} \right| > \sqrt{t^2 + 2\log p} \right) \tag{18}$$

$$= p\, \mathbb{P}\left(\left| \varepsilon' x_j / \sqrt{m\sigma^2} \right| > \sqrt{t^2 + 2\log p} \right) \quad \left(\stackrel{\text{iid}}{\sim} \mathcal{N}(0,1) \right) \tag{19}$$

- Take $V \sim \mathcal{N}(0,1)$, then for $u > 0$:

$$\mathbb{P}(V > u) = \mathbb{P}\left(e^{\lambda V} > e^{\lambda u} \right) \le \frac{\mathbb{E}(e^{\lambda V})}{e^{\lambda u}} \quad \text{by Markov's inequality } (\forall \lambda > 0) \tag{20}$$

- By knowledge of the standard normal MGF: $\mathbb{E}(e^{\lambda V}) = e^{\lambda^2/2}$, hence:

$$\mathbb{P}(V > u) \le e^{\lambda^2/2 - \lambda u} \equiv e^{f(\lambda)} \quad \text{where } f(\lambda) = \lambda^2/2 - \lambda u \tag{21}$$

- Clearly $f'(\lambda) = \lambda - u$. To minimize this function, we choose $\lambda = u$ as the inequality holds $\forall \lambda > 0$, then:

$$\mathbb{P}(V > u) \le e^{-u^2/2}. \tag{22}$$

By symmetry of the standard normal: $\mathbb{P}(V < -u) = \mathbb{P}(V > u) \le e^{-u^2/2}$.

- Hence:

$$\mathbb{P}(|V| > u) = \mathbb{P}(V > u \cup V < -u) = 2\mathbb{P}(V > u) \leq 2\exp(-u^2/2), \quad \text{so that:} \tag{23}$$

$$\mathbb{P}\left(\max_{1\leq j\leq p} \left|\varepsilon' x_j/\sqrt{m\sigma^2}\right| > \sqrt{t^2 + 2\log p}\right) \leq 2p\exp\left(-\frac{t^2 + 2\log p}{2}\right) \tag{24}$$

$$= 2p\exp(-t^2/2)\frac{1}{p} = 2\exp(-t^2/2). \tag{$**$}$$

- Hence, for $\lambda_0 = 2\sigma\sqrt{\dfrac{t^2 + 2\log p}{m}}$, we get:

$$\mathbb{P}(T) = \mathbb{P}\left(\max_{1\leq j\leq p} 2|\varepsilon' x_j|/m \leq 2\sigma\sqrt{\frac{t^2 + 2\log p}{m}}\right) \tag{25}$$

$$= \mathbb{P}\left(\max_{1\leq j\leq p} \left|\varepsilon' x_j\right| /\sqrt{\sigma^2 m} \leq \sqrt{t^2 + 2\log p}\right) \tag{26}$$

$$\Rightarrow \mathbb{P}(T) \geq 1 - 2\exp(-t^2/2) \tag{27}$$

- Consider $\hat{\sigma}$ an estimator of σ, pick the regularization parameter to be:

$$\lambda = 4\hat{\sigma}\sqrt{\frac{t^2 + 2\log p}{m}} \quad \text{s.t. } \lambda \geq 2\lambda_0 \quad \text{(arbitrary assumption)} \tag{28}$$

- From ($**$), ($*$), we get on the set T:

$$\frac{\left\|X\left(\hat{\boldsymbol{\beta}}^{\text{LASSO}} - \beta^0\right)\right\|_2^2}{m} + \lambda\left\|\hat{\boldsymbol{\beta}}^{\text{LASSO}}\right\|_1 \leq \max_{1\leq j\leq p} \frac{2|\varepsilon' x_j|}{m}\left\|\hat{\boldsymbol{\beta}}^{\text{LASSO}} - \beta^0\right\|_1 + \lambda\left\|\beta^0\right\|_1 \tag{29}$$

$$\leq \lambda_0\left\|\hat{\boldsymbol{\beta}}^{\text{LASSO}} - \beta^0\right\|_1 + \lambda\left\|\beta^0\right\|_1 \tag{30}$$

$$\leq \frac{\lambda}{2}\left\|\hat{\boldsymbol{\beta}}^{\text{LASSO}} - \beta^0\right\|_1 + \lambda\left\|\beta^0\right\|_1 \tag{31}$$

$$\iff \frac{\left\|X\left(\hat{\boldsymbol{\beta}}^{\text{LASSO}}-\beta^0\right)\right\|_2^2}{m} \le \frac{\lambda}{2}\left\|\hat{\boldsymbol{\beta}}^{\text{LASSO}}-\beta^0\right\|_1 + \lambda\left(\left\|\beta^0\right\|_1 - \left\|\hat{\boldsymbol{\beta}}^{\text{LASSO}}\right\|_1\right) \tag{32}$$

$$\le \frac{3\lambda}{2}\left\|\hat{\boldsymbol{\beta}}^{\text{LASSO}}-\beta^0\right\|_1 \tag{Δ}$$

$$\Rightarrow 2\frac{\left\|X\left(\hat{\boldsymbol{\beta}}^{\text{LASSO}}-\beta^0\right)\right\|_2^2}{m} \le 3\lambda\left\|\hat{\boldsymbol{\beta}}^{\text{LASSO}}-\beta^0\right\|_1 \tag{$***$}$$

- As $\lambda_0 = 2\sigma\sqrt{\frac{t^2+2\log p}{m}}$, $\lambda \ge 2\lambda_0 \iff \hat{\sigma} \ge \sigma$. Hence the bound $(***)$ holds on $T \cap \{\hat{\sigma} \ge \sigma\}$, which holds with probability:

$$\mathbb{P}(T \cap \{\hat{\sigma} \ge \sigma\}) \ge \mathbb{P}(T) + \mathbb{P}(\hat{\sigma} \ge \sigma) - 1 \tag{33}$$

$$\ge 1 - \left(2\exp(-t^2/2) + \mathbb{P}(\hat{\sigma} \le \sigma)\right) \equiv 1-\alpha. \tag{34}$$

- This means that, as $3\lambda\left\|\beta^0\right\|_1 \xrightarrow{m\to\infty} 0$ is needed for the LASSO to be consistent.

- Given our chosen $\lambda = 4\hat{\sigma}\sqrt{\frac{t^2+2\log p}{m}} \equiv O\left(\sqrt{\log p/m}\right)$, we obtained the bound for prediction error:

$$O\left(\left\|\beta^0\right\|_1\sqrt{\log p/m}\right), \quad \text{hence for it to vanish we require:} \tag{35}$$

$$\left\|\beta^0\right\|_1 = O\left(\sqrt{m/\log p}\right). \tag{36}$$

Allowing p to be larger than m as long as sparsity s_0 is smaller than $\sqrt{m}$.

- $\sqrt{\log p}$ is the price we pay for not knowing which variables are relevant.

- For $\lambda \ge 2\lambda_0$ to hold we need to pick $\hat{\sigma}$ carefully. For instance:

$$\hat{\sigma}^2 = \underline{y}'\underline{y}/m, \quad \text{where } \underline{y} \text{ is centered can do the job.} \tag{37}$$

- For an index set $S \subset \{1, \ldots, p\}$, define:

$$\beta_{j,S} = \beta_j \mathbb{1}(j \in S), \quad \text{so that} \quad \beta_{j,S} + \beta_{j,S^c} = \beta_j. \tag{38}$$

hence by $(***)$ on T, with $\lambda \geq 2\lambda_0$:

$$2\frac{\left\| X\left(\hat{\boldsymbol{\beta}}^{\text{LASSO}} - \beta^0\right)\right\|_2^2}{m} + 2\lambda \left\|\hat{\boldsymbol{\beta}}^{\text{LASSO}}\right\|_1 \leq \lambda \left\|\hat{\boldsymbol{\beta}}^{\text{LASSO}} - \beta^0\right\|_1 + 2\lambda \left\|\beta^0\right\|_1 \tag{39}$$

but $\left\|\hat{\boldsymbol{\beta}}\right\|_1 = \left\|\hat{\beta}_{S_0}\right\|_1 + \left\|\hat{\beta}_{S_0^c}\right\|_1 = \left\|\hat{\beta}_{S_0} - \beta^0_{S_0} + \beta^0_{S_0}\right\|_1 + \left\|\hat{\beta}_{S_0^c}\right\|_1$

$$\overset{(\Delta)}{\geq} \left\|\beta^0_{S_0}\right\|_1 - \left\|\hat{\beta}_{S_0} - \beta^0_{S_0}\right\|_1 + \left\|\hat{\beta}_{S_0^c}\right\|_1, \quad \text{and:} \tag{40}$$

$$\left\|\hat{\beta} - \beta^0\right\|_1 = \left\|\hat{\beta}_{S_0} - \beta^0_{S_0}\right\|_1 + \left\|\hat{\beta}_{S_0^c} - \beta^0_{S_0^c}\right\|_1 = \left\|\hat{\beta}_{S_0} - \beta^0_{S_0}\right\|_1 + \left\|\hat{\beta}_{S_0^c}\right\|_1 \tag{41}$$

we then obtain:

$$2\frac{\left\| X\left(\hat{\boldsymbol{\beta}}^{\text{LASSO}} - \beta^0\right)\right\|_2^2}{m} + 2\lambda \left\|\beta^0_{S_0}\right\|_1 - 2\lambda \left\|\hat{\beta}_{S_0} - \beta^0_{S_0}\right\|_1 + 2\lambda \left\|\hat{\beta}_{S_0^c}\right\|_1 \leq \lambda \left\|\hat{\beta}_{S_0} - \beta^0_{S_0}\right\|_1 + \lambda \left\|\hat{\beta}_{S_0^c}\right\|_1 + 2\lambda \left\|\beta^0_{S_0}\right\|_1 \tag{42}$$

$$\iff 2\frac{\left\| X\left(\hat{\boldsymbol{\beta}}^{\text{LASSO}} - \beta^0\right)\right\|_2^2}{m} + \lambda \left\|\hat{\beta}_{S_0^c}\right\|_1 \leq 3\lambda \left\|\hat{\beta}^{\text{LASSO}}_{S_0} - \beta^0_{S_0}\right\|_1 \tag{****}$$

- By Cauchy–Schwarz: $\left\|\hat{\beta}^{\text{LASSO}}_{S_0} - \beta^0_{S_0}\right\|_1 \leq \sqrt{\sum_{j \in S_0} |1|^2} \sqrt{\sum_{j \in S_0} \left(\hat{\beta}^{\text{LASSO}}_{S_0,j} - \beta^0_{S_0,j}\right)^2}$

$$= \sqrt{s_0} \left\|\hat{\beta}^{\text{LASSO}}_{S_0} - \beta^0_{S_0}\right\|_2. \tag{43}$$

Also: $\left\| X\left(\hat{\boldsymbol{\beta}}^{\text{LASSO}} - \beta^0\right)\right\|_2^2 / m = \left(\hat{\boldsymbol{\beta}}^{\text{LASSO}} - \beta^0\right)' \frac{X'X}{m} \left(\hat{\boldsymbol{\beta}}^{\text{LASSO}} - \beta^0\right)$

$$= \left(\hat{\boldsymbol{\beta}}^{\text{LASSO}} - \beta^0\right)' \hat{\Sigma} \left(\hat{\boldsymbol{\beta}}^{\text{LASSO}} - \beta^0\right) \tag{44}$$

- From $(****)$, on T for $\lambda \geq 2\lambda_0$, we have:

$$\left\|\hat{\beta}^{\text{LASSO}}_{S_0^c}\right\|_1 \leq 3\left\|\hat{\beta}^{\text{LASSO}}_{S_0} - \beta^0_{S_0}\right\|_1 \tag{45}$$

- We need some conditions (compatibility conditions) on the design matrix. This condition is said to hold on S_0 if for some $\phi_0 > 0$ and $\forall\beta$ satisfying

$$\left\|\beta_{S_0^c}\right\|_1 \leq 3\left\|\beta_{S_0}\right\|_1, \quad \text{it holds that} \quad \|\beta_{S_0}\|_1^2 \leq \left(\beta'\hat{\Sigma}\beta\right) s_0/\phi_0^2. \tag{46}$$

In other words, this assumes that X does not exhibit extreme collinearity.

- Then, if we replace $\left\|\beta^0_{S_0}\right\|_1^2$ by an upper bound $s_0\left\|\hat{\beta}_{S_0}\right\|_2^2$:

$$\begin{aligned} 2\frac{\left\|X\left(\hat{\boldsymbol{\beta}} - \beta^0\right)\right\|_2^2}{m} + \lambda\left\|\hat{\boldsymbol{\beta}} - \beta^0\right\|_1 &= 2\frac{\left\|X\left(\hat{\boldsymbol{\beta}} - \beta^0\right)\right\|_2^2}{m} + \lambda\left\|\hat{\beta}_{S_0} - \beta^0_{S_0}\right\|_1 + \lambda\left\|\hat{\beta}_{S_0^c}\right\|_1 \qquad (47) \\ &\leq 4\lambda\left\|\hat{\beta}_{S_0} - \beta^0_{S_0}\right\|_1 \qquad (*****) \end{aligned}$$

- Because $\left\|\hat{\beta}_{S_0^c}\right\|_1 \leq 3\left\|\hat{\beta}_{S_0} - \beta^0_{S_0}\right\|_1$ and we assume the compatibility condition, it then holds that:

$$\begin{aligned} \left\|\hat{\beta}_{S_0} - \beta^0_{S_0}\right\|_1^2 &\leq \left(\hat{\boldsymbol{\beta}} - \beta^0\right)'\hat{\Sigma}\left(\hat{\boldsymbol{\beta}} - \beta^0\right) s_0/\phi_0^2 \qquad (48) \\ &= \left\|X\left(\hat{\boldsymbol{\beta}} - \beta^0\right)\right\|_2^2 \frac{s_0}{\phi_0^2 m}, \quad \text{hence:} \qquad (49) \end{aligned}$$

$$\begin{aligned} 2\frac{\left\|X\left(\hat{\boldsymbol{\beta}} - \beta^0\right)\right\|_2^2}{m} + \lambda\left\|\hat{\boldsymbol{\beta}} - \beta^0\right\|_1 \leq 4\lambda\left\|X\left(\hat{\boldsymbol{\beta}} - \beta^0\right)\right\|_2 \frac{\sqrt{s_0}}{\phi_0\sqrt{m}} &= 4\left(\frac{\left\|X\left(\hat{\boldsymbol{\beta}} - \beta^0\right)\right\|_2}{\sqrt{m}}\right)\left(\lambda\frac{\sqrt{s_0}}{\phi_0}\right) \qquad (50) \\ &\leq \frac{\left\|X\left(\hat{\boldsymbol{\beta}} - \beta^0\right)\right\|_2^2}{m} + 4\lambda^2 s_0/\phi_0^2 \qquad (51) \end{aligned}$$

using $4uv \leq u^2 + 4v^2$, indeed by Young's inequality: $uv \leq u^2/2 + v^2/2 \iff 4uv \leq u^2 + v^2 \leq u^2 + 4v^2$.

- Then, on T, for $\lambda \geq 2\lambda_0$, we get:

$$\frac{\left\| X\left(\hat{\boldsymbol{\beta}}^{\text{LASSO}} - \beta^0\right)\right\|_2^2}{m} + \lambda \left\|\hat{\boldsymbol{\beta}}^{\text{LASSO}} - \beta^0\right\|_1 \leq 4\lambda^2 s_0/\phi_0^2 \qquad (******)$$

where the first term on the left is the *prediction* term and the second is the *sparsity* term.

- We already know that for $\lambda = 4\hat{\sigma}\sqrt{\frac{t^2 + 2\log p}{m}}$ for some $t > 0$ and assuming that $\hat{\sigma}_j^2 = 1\ \forall j$, with probability at least $1 - \left[2\exp(-t^2/2) + \mathbb{P}(\hat{\sigma} \leq \sigma)\right] \equiv \alpha$, we get $2\left\| X\left(\hat{\boldsymbol{\beta}}^{\text{LASSO}} - \beta^0\right)\right\|_2^2/m \leq 3\lambda\left\|\beta^0\right\|_1$. Because the bound $(******)$ also used the same assumption (on T, for $\lambda \geq 2\lambda_0$), it then also holds with probability at least $1 - \alpha$.

II) Results using a linear approximation of the truth

- Extend the theory to cases where $\mathbb{E}\underline{y} = f^0$ where f^0 is not necessarily a sparse linear index of the variables.
- We can easily obtain the same inequality as $(**)$, $\forall\beta^*$, accounting for the approximation error:

$$\frac{\left\| X\hat{\boldsymbol{\beta}}^{\text{LASSO}} - f^0\right\|_2^2}{m} + \lambda\left\|\hat{\boldsymbol{\beta}}^{\text{LASSO}}\right\|_1 \leq \frac{2\varepsilon' X\left(\hat{\boldsymbol{\beta}}^{\text{LASSO}} - \beta^*\right)}{m} + \lambda\left\|\beta^*\right\|_1 + \left\| X\beta^* - f^0\right\|_2^2/m. \qquad (**_f)$$

- As before, on T, now assuming that $\lambda \geq 4\lambda_0$, we get:

$$\begin{aligned}
\frac{\left\| X\hat{\boldsymbol{\beta}}^{\text{LASSO}} - f^0\right\|_2^2}{m} + \lambda\left\|\hat{\boldsymbol{\beta}}^{\text{LASSO}}\right\|_1 &\leq \max_{1\leq j\leq p}\frac{2|\varepsilon' x_j|}{m}\left\|\hat{\boldsymbol{\beta}}^{\text{LASSO}} - \beta^*\right\|_1 + \lambda\left\|\beta^*\right\|_1 && (52)\\
&\quad + \frac{\left\| X\beta^* - f^0\right\|_2^2}{m} && (53)\\
&\leq \lambda_0\left\|\hat{\boldsymbol{\beta}}^{\text{LASSO}} - \beta^*\right\|_1 + \lambda\left\|\beta^*\right\|_1 + \left\| X\beta^* - f^0\right\|_2^2/m && (54)\\
&\leq \frac{\lambda}{4}\left\|\hat{\boldsymbol{\beta}}^{\text{LASSO}} - \beta^*\right\|_1 + \lambda\left\|\beta^*\right\|_1 + \left\| X\beta^* - f^0\right\|_2^2/m && (55)
\end{aligned}$$

$$\iff \frac{\left\|X\hat{\boldsymbol{\beta}}^{\text{LASSO}}-f^0\right\|_2^2}{m} \leq \frac{\lambda}{4}\left\|\hat{\boldsymbol{\beta}}^{\text{LASSO}}-\beta^*\right\|_1+\lambda\left(\|\beta^*\|_1-\left\|\hat{\boldsymbol{\beta}}^{\text{LASSO}}\right\|_1\right)+\left\|X\beta^*-f^0\right\|_2^2/m \tag{56}$$

$$\leq \frac{5\lambda}{4}\left\|\hat{\boldsymbol{\beta}}^{\text{LASSO}}-\beta^*\right\|_1+\left\|X\beta^*-f^0\right\|_2^2/m \tag{$**_f\ \rho$}$$

- Let $S_* = \{j : \beta_j^* \neq 0\}$, then:

$$\left\|\hat{\boldsymbol{\beta}}^{\text{LASSO}}\right\|_1 = \left\|\hat{\beta}_{S_*}\right\|_1+\left\|\hat{\beta}_{S_*^c}\right\|_1 \tag{57}$$

- As $\left\|\hat{\beta}_{S_*}+\hat{\beta}_{S_*^c}\right\|_1 \geq \|\beta^*_{S_*}\|_1-\left\|\hat{\beta}^{\text{LASSO}}_{S_*}-\beta^*_{S_*}\right\|_1+\left\|\hat{\beta}^{\text{LASSO}}_{S_*^c}\right\|_1$, and:

$$\left\|\hat{\boldsymbol{\beta}}^{\text{LASSO}}-\beta^*\right\|_1 = \left\|\hat{\beta}^{\text{LASSO}}_{S_*}-\beta^*_{S_*}\right\|_1+\left\|\hat{\beta}^{\text{LASSO}}_{S_*^c}\right\|_1, \quad \text{so that } (**_f\ \rho) \iff \tag{58}$$

$$4\frac{\left\|X\hat{\boldsymbol{\beta}}^{\text{LASSO}}-f^0\right\|_2^2}{m}+4\lambda\left\|\hat{\boldsymbol{\beta}}^{\text{LASSO}}\right\|_1 \leq \lambda\left\|\hat{\beta}^{\text{LASSO}}_{S_*}-\beta^*\right\|_1+4\lambda\|\beta^*\|_1+4\left\|X\beta^*-f^0\right\|_2^2/m \tag{59}$$

$$\begin{aligned}\Rightarrow 4\frac{\left\|X\hat{\boldsymbol{\beta}}^{\text{LASSO}}-f^0\right\|_2^2}{m}&+4\lambda\left\|\hat{\beta}^{\text{LASSO}}_{S_*}\right\|_1-4\lambda\left\|\hat{\beta}^{\text{LASSO}}_{S_*}-\beta^*_{S_*}\right\|_1+4\lambda\left\|\hat{\beta}^{\text{LASSO}}_{S_*^c}\right\|_1\\ &\leq \lambda\left\|\hat{\beta}^{\text{LASSO}}_{S_*}-\beta^*_{S_*}\right\|_1+\lambda\left\|\hat{\beta}^{\text{LASSO}}_{S_*^c}\right\|_1+4\lambda\left\|\beta^*_{S_*}\right\|_1+4\left\|X\beta^*-f^0\right\|_2^2/m.\end{aligned} \tag{60}$$

$$\iff 4\frac{\left\|X\hat{\boldsymbol{\beta}}^{\text{LASSO}}-f^0\right\|_2^2}{m}+3\lambda\left\|\hat{\beta}^{\text{LASSO}}_{S_*^c}\right\|_1 \leq 5\lambda\left\|\hat{\beta}^{\text{LASSO}}_{S_*}-\beta^*_{S_*}\right\|_1+4\left\|X\beta^*-f^0\right\|_2^2/m \tag{$**_f\ \rho\rho$}$$

- Contrary to $(****)$, $(**_f\,\rho\rho)$ does not allow us to conclude that:

$$\left\|\hat{\beta}^{\text{LASSO}}_{S_*^c}\right\|_1 \leq \frac{5}{3}\left\|\hat{\beta}^{\text{LASSO}}_{S_*} - \beta^*_{S_*}\right\|_1 \leq 3\left\|\hat{\beta}^{\text{LASSO}}_{S_*} - \beta^*_{S_*}\right\|_1 \quad \text{because of the approximation error.} \tag{61}$$

- There are two cases:
  (i) $\lambda\left\|\hat{\beta}_{S_*} - \beta^*_{S_*}\right\|_1 \geq \left\|X\beta^* - f^0\right\|_2^2/m$
  (ii) $\lambda\left\|\hat{\beta}_{S_*} - \beta^*_{S_*}\right\|_1 \leq \left\|X\beta^* - f^0\right\|_2^2/m$

$$\iff \text{(i)} \quad 4\frac{\left\|X\hat{\beta} - f^0\right\|_2^2}{m} + 3\lambda\left\|\hat{\beta}_{S_*^c}\right\|_1 \leq 9\lambda\left\|\hat{\beta}_{S_*} - \beta^*_{S_*}\right\|_1 \tag{62}$$

$$\text{(ii)} \quad 4\frac{\left\|X\hat{\beta} - f^0\right\|_2^2}{m} + 3\lambda\left\|\hat{\beta}_{S_*^c}\right\|_1 \leq 9\frac{\left\|X\beta^* - f^0\right\|_2^2}{m} \tag{63}$$

- The compatibility condition can be generalized. It is said to hold for a given set S if $\exists$ some $\phi(S) > 0$ and $\forall\beta$ satisfying $\|\beta_{S^c}\|_1 \leq 3\|\beta_S\|_1$, we have:

$$\|\beta_S\|_1^2 \leq \left(\beta'\hat{\Sigma}\beta\right)|S|/\phi^2(S) \tag{64}$$

- Let $\mathcal{S}$ a collection of index sets s.t. the compatibility condition holds.
- The oracle is $\beta^* = \arg\min_{\beta: S_\beta \in \mathcal{S}}\left\{\left\|X\beta - f^0\right\|_2^2/m + \underbrace{4\lambda^2 s_\beta/\phi^2(S_\beta)}\right\}$

  adding variables reduces bias, but increases variance.

- The point is that this ideal depends on unobserved quantities: f^0, σ^2 (through λ) and $\phi(S)$, yet under our assumptions LASSO is able to achieve the same performance up to a constant: this is the oracle property.
- Given S, we first look for the best approximation of f^0 using only non-zero coefficients in that set:

$$b^S = \arg\min_{\beta=\beta_S} \left\{ \left\| X\beta - f^0 \right\|_2^2 \right\}, \quad \text{and note } f_S = Xb^S. \tag{65}$$

- Then, we minimize over all $S \in \mathcal{S}$ with an ℓ_0 penalty:

$$S_* = \arg\min_{S\in\mathcal{S}} \left\{ \left\| f_S - f^0 \right\|_2^2 / m + \frac{4\lambda^2 |S|}{\phi^2(S)} \right\} \tag{66}$$

- Then, the oracle is $\beta^* = b^{S_*}$.
- The probability to be on T is still at least $1 - \left[2\exp(-t^2/2) + \mathbb{P}(\hat{\sigma} \leq \sigma)\right]$ (and that $\hat{\sigma} > \sigma$).
- Consider $\lambda = 8\hat{\sigma}\sqrt{\dfrac{t^2 + 2\log p}{m}}$.
- For case (i), we have:

$$4\frac{\left\| X\hat{\beta} - f^0 \right\|_2^2}{m} + 3\lambda \left\| \hat{\beta}_{S_*^c} \right\|_1 \leq 9\lambda \left\| \hat{\beta}_{S_*} - \beta^*_{S_*} \right\|_1 \tag{67}$$

$$\Rightarrow \left\| \hat{\beta}_{S_*^c} \right\| \leq 3 \left\| \hat{\beta}_{S_*} - \beta^*_{S_*} \right\|_1, \quad \text{then by the generalized compatibility condition, we get:} \tag{68}$$

$$\left\| \hat{\beta}_{S_*} - \beta^*_{S_*} \right\|_1^2 \leq \left(\hat{\boldsymbol{\beta}} - \beta^* \right)' \hat{\Sigma} \left(\hat{\boldsymbol{\beta}} - \beta^* \right) |S_*| / \phi^2(S_*) \tag{69}$$

$$= \left\| X \left(\hat{\boldsymbol{\beta}} - \beta^* \right) \right\|_2^2 |S_*| / \left(\phi^2(S_*)\, m \right) \tag{70}$$

- So that:

$$4\frac{\left\|X\hat{\beta}-f^0\right\|_2^2}{m}+3\lambda\left\|\hat{\beta}_{S_*^c}\right\|_1+3\lambda\left\|\hat{\beta}_{S_*}-\beta^*_{S_*}\right\|_1\leq 12\lambda\left\|\hat{\beta}_{S_*}-\beta^*_{S_*}\right\|_1 \tag{71}$$

as $\left\|\hat{\beta}-\beta^*\right\|_1=\left\|\hat{\beta}_{S_*}-\beta^*_{S_*}\right\|_1+\left\|\hat{\beta}_{S_*^c}\right\|_1$, we get:

$$4\frac{\left\|X\hat{\beta}-f^0\right\|_2^2}{m}+3\lambda\left\|\hat{\beta}-\beta^*\right\|_1\leq 12\lambda\left\|\hat{\beta}_{S_*}-\beta^*_{S_*}\right\|_1 \tag{72}$$

$$\leq 12\lambda\left\|X\left(\hat{\beta}-\beta^*\right)\right\|_2\frac{\sqrt{s_*}}{\sqrt{m}\,\phi(S_*)} \tag{73}$$

By Young's inequality: $uv\leq u^2/2+v^2/2\Rightarrow 12uv\leq 6u^2+6v^2$. Similarly, pick $a=6u$ and $b=2v$, then: $ab\leq a^2/2+b^2/2 \iff 12uv\leq 18u^2+2v^2$.

$$\leq 12\left(\frac{\lambda\sqrt{s_*}}{\phi_*}\right)\left(\frac{\left\|X\left(\hat{\beta}-\beta^*\right)\right\|_2}{\sqrt{m}}\right) \tag{74}$$

$$\leq 12\frac{\lambda\sqrt{s_*}}{\phi_*}\left(\frac{\left\|X\hat{\beta}-f^0\right\|_2+\left\|f^0-X\beta^*\right\|_2}{\sqrt{m}}\right) \tag{Δ}$$

$$=12\left(\frac{\lambda\sqrt{s_*}}{\phi_*}\right)\left(\frac{\left\|X\hat{\beta}-f^0\right\|_2}{\sqrt{m}}\right)+12\left(\frac{\lambda\sqrt{s_*}}{\phi_*}\right)\left(\frac{\left\|X\beta^*-f^0\right\|_2}{\sqrt{m}}\right) \tag{75}$$

$$\leq 18\frac{\lambda^2 s_*}{\phi_*^2}+2\frac{\left\|X\hat{\beta}-f^0\right\|_2^2}{m}+6\frac{\lambda^2 s_*}{\phi_*^2}+6\frac{\left\|X\beta^*-f^0\right\|_2^2}{m} \tag{76}$$

$$=24\frac{\lambda^2 s_*}{\phi_*^2}+2\frac{\left\|X\hat{\beta}-f^0\right\|_2^2}{m}+6\frac{\left\|X\beta^*-f^0\right\|_2^2}{m} \tag{77}$$

$$\Rightarrow 2\frac{\left\|X\hat{\beta}-f^0\right\|_2^2}{m}+3\lambda\left\|\hat{\beta}-\beta^*\right\|_1\leq 24\frac{\lambda^2 s_*}{\phi_*^2}+6\frac{\left\|X\beta^*-f^0\right\|_2^2}{m} \tag{78}$$

- For case (ii), we get:

$$4\frac{\left\|X\hat{\beta}-f^0\right\|_2^2}{m}+3\lambda\left\|\hat{\beta}_{S_*^c}\right\|_1\leq 9\frac{\left\|X\beta^*-f^0\right\|_2^2}{m} \tag{79}$$

$$\Rightarrow 4\frac{\left\|X\hat{\beta}-f^0\right\|_2^2}{m}+3\lambda\left(\left\|\hat{\beta}_{S_*}-\beta^*_{S_*}\right\|_1+\left\|\hat{\beta}_{S_*^c}\right\|_1\right)\leq 9\frac{\left\|X\beta^*-f^0\right\|_2^2}{m}+3\lambda\left\|\hat{\beta}_{S_*}-\beta^*_{S_*}\right\|_1 \tag{80}$$

$$\Rightarrow 4\frac{\left\|X\hat{\beta}-f^0\right\|_2^2}{m}+3\lambda\left\|\hat{\beta}-\beta^*\right\|_1\leq 12\frac{\left\|X\beta^*-f^0\right\|_2^2}{m} \tag{81}$$

- Hence, for that value of λ, we have with probability at least $1-\alpha$:

$$\underbrace{2\frac{\left\|X\hat{\beta}-f^0\right\|_2^2}{m}}_{\text{prediction error}}+\underbrace{\lambda\left\|\hat{\beta}-\beta^*\right\|_1}_{\substack{\text{distance to}\\\text{best sparse linear proxy}}}\leq\underbrace{6\frac{\left\|X\beta^*-f^0\right\|_2^2}{m}}_{\text{approximation error}}+\underbrace{\frac{24\lambda^2 s_*}{\phi_*^2}}_{\substack{\text{statistical cost of}\\\text{estimating the active coefficients}}} \tag{82}$$

- Hence, the LASSO still works well as long as the true relationship can be approximated by some sparse linear model.